Mathematical Foundations (Continued)

★ Matrix :-

A rectangular arrangement of numbers in the form of rows & columns.

Ex- $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \\ -5 & 8 \end{bmatrix}$

Column ↑ (3 columns)

→ row (3 rows)

m = no. of rows
n = no. of columns

m×n → order of matrix

Types of Matrix :-

(1) Row Matrix ⇝ $[a_1 \quad a_2 \quad a_3 \quad \dots]$

(2) Column Matrix ⇝ $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \end{bmatrix}$

(3) Square Matrix ⇝ $\begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix}$

(4) Zero Matrix ⇝ $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(5) Identity Matrix ⇝ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

★ Basic Matrix Operations :-

(1) Addition of matrix

(2) Subtraction of matrix

(2) Subtraction of matrix

(3) Multiplication " "

(4) Transpose

(5) Determinant

* Matrix Multiplication :-

I. Two matrix are selected such that no. of columns of one matrix is equal to no. of rows of another matrix.

Ex- $A = \begin{bmatrix} 2 & 5 & 1 \\ 4 & 0 & 7 \end{bmatrix}_{2\times 3}$, $B = \begin{bmatrix} 1 & 0 & 5 \\ 2 & 6 & 3 \\ 7 & 5 & 8 \end{bmatrix}_{3\times 3}$

$$A \cdot B = \begin{bmatrix} 2\times1+5\times2+1\times7 & 2\times0+5\times6+1\times5 & 2\times5+5\times3+1\times8 \\ 4\times1+0\times2+7\times7 & 4\times0+0\times6+7\times5 & 4\times5+0\times3+7\times8 \end{bmatrix}$$

$$A \cdot B = \begin{bmatrix} 19 & 35 & 33 \\ 53 & 35 & 76 \end{bmatrix}_{2\times3}$$

II. $A \cdot B \neq B \cdot A$

* Transpose of a Matrix :-

↳ Changing rows into columns & vice versa is known as the transpose of a matrix

Ex- $A = \begin{bmatrix} 2 & 4 & 7 \\ 3 & 5 & 1 \end{bmatrix}$ → Attn. $A^T = \begin{bmatrix} 2 & 3 \\ \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 4 & 7 \\ 3 & 5 & 1 \end{bmatrix} \xrightarrow{\text{After transposing}} A^T = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 7 & 1 \end{bmatrix}$$

* Dot Product of a Vector :-

↳ Suppose we have two vectors $\vec{A}$ & $\vec{B}$

$$\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k} \quad , \quad \vec{B} = 4\hat{i} + 5\hat{j} + 3\hat{k}$$

$$\boxed{\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos\theta}$$

where, $|A|$ = Magnitude of $\vec{A}$

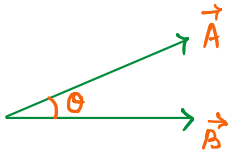
$|B|$ = " of $\vec{B}$

$\cos\theta$ = $\angle$ b/w $\vec{A}$ & $\vec{B}$

$$|\vec{A}| = \sqrt{(2i)^2 + (3j)^2 + (4k)^2}$$

$$|\vec{A}| = \sqrt{4 + 9 + 16}$$

$$|\vec{A}| = \sqrt{29}$$



or

$$\boxed{\cos\theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}|\,|\vec{B}|}}$$

Above we studied i, j, K ⇝ components of x, y, z axis

$$i \cdot i = 1 \quad , \quad i \cdot j = 0 \quad , \quad j \cdot i = 0$$

$$j \cdot j = 1 \quad , \quad j \cdot k = 0 \quad , \quad k \cdot j = 0$$

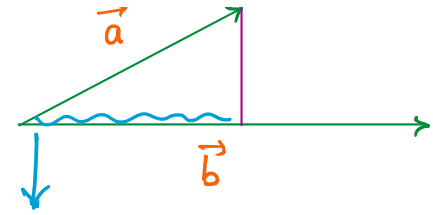
$$k \cdot k = 1 \quad , \quad i \cdot K = 0 \quad , \quad K \cdot i = 0$$

* Projection of a Vector :-

Suppose,







Projection of A on B

Formula →

Projection of $\vec{a}$ on $\vec{b}$:

$$\text{Proj}_{\vec{b}}\ \vec{a} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\cdot\vec{b}$$

★ Calculus :-

↳ Branch of Mathematics where we observe continuous change

- **Differentiation :-**

$\frac{d}{dx}$ ⇝ general representation of diff.

Formula :-

- $\frac{d}{dx}x^n = nx^{n-1}$

Ex - $\frac{d}{dx}x^2 = 2x^{2-1} = 2x$

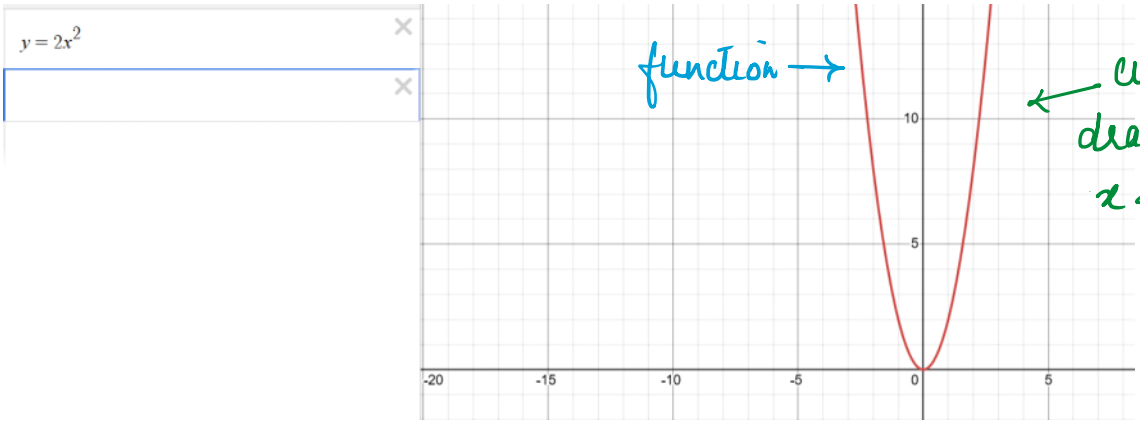
- $\frac{d}{dx}y^3 = 3y^2 \cdot \frac{dy}{dx}$

★ Maxima & Minima of a function :-

* Maxima & Minima of a function :-

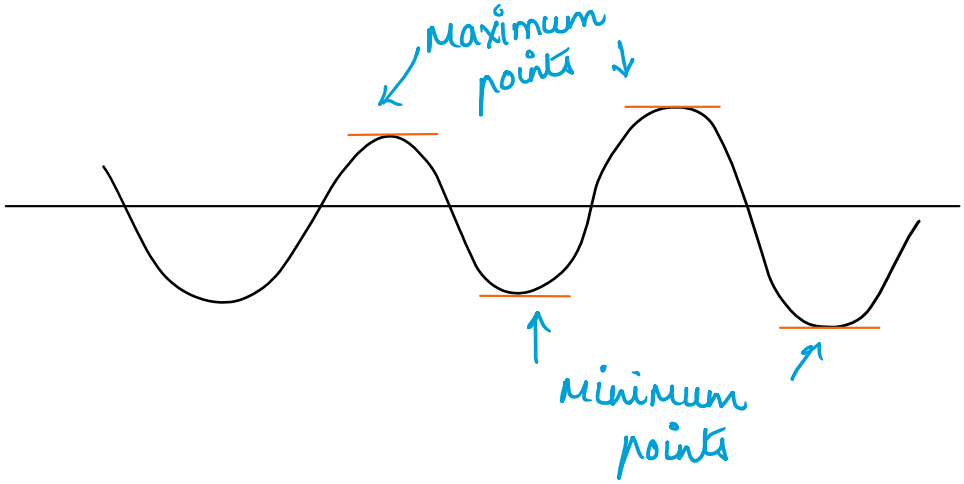
function :- A mathematical rule that takes an i/p, processes it & produces an o/p.

Ex- $y = 2x^2$ or $F(x) = 2x^2$



As we can see that a function can be represented graphically

↳ There can be certain ups & downs in that curve



These maximum & minimum values on the graph or curve are known as maxima & minima respectively.

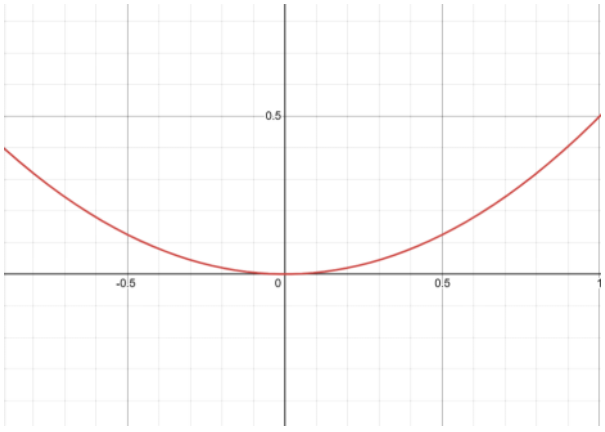
* Continuous & Discontinuous functions :-

↓

A function with no break

Ex- square function

↓

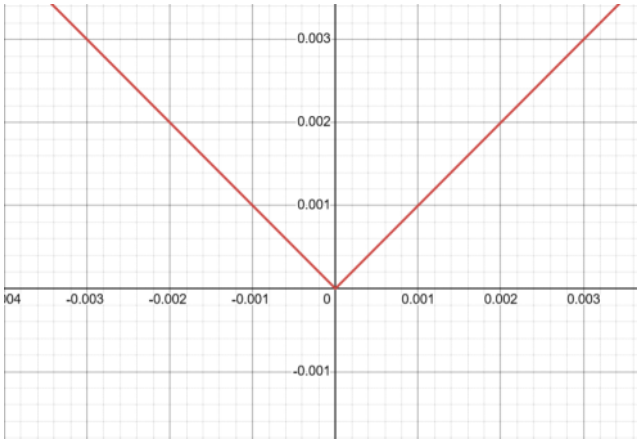


↓

A function which breaks in b/w

Ex- (1) Modulus function

↓



(2) Step function

